

The boundary tail of the standard integral is exponentially small

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Abstract

Third §4.2 foothold: the one-dimensional population version of the remainder estimate `eq:Deltaupper`. The paper's standard integral lives on the chart $[0, b]$; the fluctuation-function identity of unit 22 lives on $(0, \infty)$. We bound the difference — the tail over (b, ∞) — by $e^{\beta a^2/2} e^{-(\beta/4)nb^{2k}}$ times an explicit Gamma constant, for $n \geq 1$, and conclude that the chart integral equals $(2k)^{-1} n^{-(h+1)/(2k)} S_{(h+1)/(2k)}(a)$ up to $O(e^{-\varepsilon n})$ with $\varepsilon = (\beta/4)b^{2k}$. Zero sorry/axiom.

1 Setting

In the proof of `thm:TaylorTree` the state-density integral is cut at $T = nb^{2|k|}$ and the remainder Δ is bounded by $e^{-\beta T/4}$ times a convergent multi-sum. Stripped of the tree combinatorics, the analytic content is: on $u \geq b$ the exponent $-\beta nu^{2k} + \beta \sqrt{n} u^k a$ is at most $-\frac{\beta}{4}nb^{2k} - \frac{\beta}{4}u^{2k} + \frac{\beta}{2}a^2$, so the tail is dominated by a fixed Gamma integral times $e^{-(\beta/4)nb^{2k}}$.

2 The things formalised

```
theorem standardIntegrand_le ... :
  Real.exp (- *n*u^(2*k) + *Real.sqrt n*u^k*a)
    Real.exp (*a^2/2) * Real.exp (-(/2*n) * u^(2*k))
theorem standardIntegrand_integrableOn ... :
  IntegrableOn (fun u => u^h * exp(...)) (Ioi 0)
theorem standardIntegral1D_tail_le ( n a b : ) (h k : ) (h : 0 < ) (hn : 1 n)
  (hb : 0 < b) (hk : 0 < k) :
  ( u in Ioi b, u^h * exp(...))
    exp(*a^2/2) * exp(-(/4)*n*b^(2*k))
    * ((/4)^(-((h:)+1)/(2*k))) * (1/(2*(k:))) * Gamma(((h:)+1)/(2*k)))
theorem standardIntegral1D_bounded_approx ... :
  |( u in Ioc 0 b, u^h * exp(...))
    - (1/(2*(k:))) * n^(-((h:)+1)/(2*k))) * fluctuation (((h:)+1)/(2*k)) a|
  (the same bound)
```

3 Proof strategy

`standardIntegrand_le` is the Gaussian domination of unit 23 evaluated at $t = nu^{2k}$, using $\sqrt{nu^{2k}} = \sqrt{n} u^k$. Integrability on $(0, \infty)$ follows by `Integrable.mono'` against $e^{\beta a^2/2} u^h e^{-(\beta/2)nu^{2k}}$, which is `integrableOn_rpow_mul_exp_neg_mul_rpow` after `rpow_natCast`. For the tail, on $u > b$ we have $u^{2k} \geq b^{2k}$ and $n \geq 1$, so $-\frac{\beta}{2}nu^{2k} \leq -\frac{\beta}{4}nb^{2k} - \frac{\beta}{4}u^{2k}$ (`nlinarith` with the two product hints).

$\beta n(u^{2k} - b^{2k}) \geq 0$ and $\beta(n-1)u^{2k} \geq 0$); the tail integral is then bounded by `setIntegral_mono_on` against $C \cdot u^h e^{-(\beta/4)u^{2k}}$, enlarged from (b, ∞) to $(0, \infty)$ by `setIntegral_mono_set`, and evaluated by `integral_rpow_mul_exp_neg_mul_rpow`. The chart approximation splits $(0, \infty)$ as $(0, b] \cup (b, \infty)$ (`Ioc_union_Ioi_eq_Ioi`, `setIntegral_union`) and invokes unit 22 for the full-line value.

4 Roadblocks and resolutions

- `Integrable.const_mul` returns a bare `Integrable` (an `And`), so dot-notation `.mono_set` fails with “`And.mono_set`”; bind it first with the `IntegrableOn` type ascribed.
- `pow_le_pow_left` is now `pow_le_pow_left` on this pin.
- The a.e. nonnegativity hypothesis of `setIntegral_mono_set` is stated with `[.restrict t]; unfold Filter.EventuallyLE then ae_restrict_iff'` to reduce it to a pointwise statement on the set, and use `HasSubset.Subset.eventuallyLE` for the set inclusion.
- The Gamma-lemma output carries $-(h+1)/(2k)$; `push_cast` then `neg_div` aligns it with the stated $-((h+1)/(2k))$.

5 What was Mathlib, what was new

Mathlib: the Gamma-integral pair, `Integrable.mono'`, `setIntegral_mono_on`, `setIntegral_mono_set`, `setIntegral_union`, `Ioc_union_Ioi_eq_Ioi`. New: integrability of the standard integrand, the explicit exponential tail bound with its Gamma constant, and the chart-vs-half-line approximation — the 1D skeleton of $\Delta = O(e^{-\varepsilon n})$.

6 Lessons

A domination lemma factored out once (unit 23) pays twice: it gives integrability and the tail bound with the same majorant. State tail bounds with the explicit constant rather than an $O(\cdot)$ — the constant is one Gamma evaluation away and makes the result quotable.

7 Follow-ups

The 1D population Taylor-tree pipeline now has all its analytic steps (kernel identity, insertions, tail). What remains multivariate: the tree-coefficient series interchange and the state-density Mellin machinery.